Unit 3 - Session 9

SLO 1: Applications & Review of HT/GHT- Assess HT/GHT's role in modern vision pipelines

Duration: 10 Minutes Activity

Question:

INSTRUCTIONS

The diagram below shows a standard modern computer vision pipeline with 6 stages. For each stage you must:

1. State whether HT/GHT is applicable at this stage (YES or NO)
2. Describe what HT/GHT does at this stage (or why it cannot be applied)
3. Name a deep learning (DL) alternative that performs the same function at this stage
4. Judge: is HT/GHT better, worse, or comparable to DL at this stage — and why?

THE PIPELINE — Study the Flow Before Filling in the Cards

The diagram below shows the 6 stages of a modern computer vision pipeline. Study the flow, then complete one card per stage.



PART A — Stage-by-Stage Analysis (Complete one card per stage)

For each stage: circle YES or NO for Q1, then fill in Q2, Q3 and Q4 in the answer lines provided.

S1 IMAGE INPUT

STAGE 1 — IMAGE INPUT

The raw image enters the pipeline from a camera, file, or sensor. This stage covers image acquisition, colour space, and resolution.

[Hint] *Think about whether GHT operates on raw pixels or requires any prior processing.*

Q1. Is HT/GHT applicable here? [YES / NO] (circle one) Answer: NO

Q2. What does HT/GHT do at this stage? (or why is it not applicable?)

HT/GHT cannot operate on raw pixels; it requires processed boundary or edge features.

Q3. Name a Deep Learning alternative that performs the same function here:

CNN-based image preprocessing

Q4. At this stage, HT/GHT is [BETTER / WORSE / COMPARABLE] than DL. Reason:

WORSE — HT/GHT does not perform image acquisition or raw-pixel preparation.

S2
PRE-PROCESSING

STAGE 2 — PRE-PROCESSING

The image is prepared for analysis: noise reduction (Gaussian blur), contrast enhancement, colour-to-grayscale conversion, and edge detection (Canny/Sobel).

[Hint] *HT requires edge maps as input. Consider which pre-processing steps are prerequisites for HT.*

Q1. Is HT/GHT applicable here? [YES / NO] (circle one) Answer: NO

Q2. What does HT/GHT do at this stage? (or why is it not applicable?)

HT/GHT does not preprocess; it uses the edge map produced here as input.

Q3. Name a Deep Learning alternative that performs the same function here:

CNN-based denoising / learned preprocessing

Q4. At this stage, HT/GHT is [BETTER / WORSE / COMPARABLE] than DL.

Reason:

WORSE — DL can learn preprocessing, while HT/GHT requires an explicit edge map.

S3
FEATURE EXTRACTION

STAGE 3 — FEATURE EXTRACTION

Meaningful patterns are extracted from the image: edges, gradients, keypoints, corner responses, or learned feature maps from a neural network.

[Hint] *GHT builds its R-table from boundary features. Where does feature extraction sit relative to the voting stage?*

Q1. Is HT/GHT applicable here? [YES / NO] (circle one) Answer: YES

Q2. What does HT/GHT do at this stage? (or why is it not applicable?)

GHT uses boundary features and edge orientations to build or query the R-table before voting.

Q3. Name a Deep Learning alternative that performs the same function here:

CNN feature extractor

Q4. At this stage, HT/GHT is [BETTER / WORSE / COMPARABLE] than DL.

Reason:

WORSE — learned features adapt better to varied appearances than hand-crafted boundary features.

S4
DETECTION

STAGE 4 — DETECTION

Objects, shapes, or regions of interest are located in the image. This is where voting, classification, or regression produces candidate detections.

[Hint] *This is the core stage for HT/GHT. The accumulator peak search happens here.*

Q1. Is HT/GHT applicable here? [YES / NO] (circle one) Answer: YES

Q2. What does HT/GHT do at this stage? (or why is it not applicable?)

HT/GHT votes in parameter space; accumulator peaks give the detected object location and shape.

Q3. Name a Deep Learning alternative that performs the same function here:

YOLO object detector

Q4. At this stage, HT/GHT is [BETTER / WORSE / COMPARABLE] than DL.

Reason:

WORSE — DL is generally more robust to appearance variation and clutter.

S5 POST-PROCESSING

STAGE 5 — POST-PROCESSING

Raw detections are refined: non-maximum suppression, false positive filtering, bounding box regression, confidence thresholding, or result smoothing across frames.

[Hint] *Think about whether HT produces clean outputs directly or whether further processing is needed.*

Q1. Is HT/GHT applicable here? [YES / NO] (circle one) Answer: YES

Q2. What does HT/GHT do at this stage? (or why is it not applicable?)

HT/GHT uses peak selection and thresholding to refine raw candidate detections.

Q3. Name a Deep Learning alternative that performs the same function here:

Non-maximum suppression and confidence filtering

Q4. At this stage, HT/GHT is [BETTER / WORSE / COMPARABLE] than DL.

Reason:

COMPARABLE — both refine candidate detections, although HT/GHT may need additional filtering.

S6 OUTPUT

STAGE 6 — OUTPUT

The final result is delivered: annotated image, parameter values (cx, cy, a, b, theta), decision (match/no-match), or structured data for downstream use.

[Hint] *Consider what form HT outputs its result and whether this format suits downstream applications.*

Q1. Is HT/GHT applicable here? [YES / NO] (circle one) Answer: NO

Q2. What does HT/GHT do at this stage? (or why is it not applicable?)

HT/GHT produces detection parameters or decisions; delivering the final result is not an HT/GHT operation.

Q3. Name a Deep Learning alternative that performs the same function here:

Detector prediction/output layer

Q4. At this stage, HT/GHT is [BETTER / WORSE / COMPARABLE] than DL. Reason:

COMPARABLE — both can return structured results for downstream use.

PART B — Summary Table (Transfer your answers from Part A)

Fill in this summary table using your detailed answers from Part A. Keep each entry brief — one phrase or one word is enough.

Stage	HT/GHT Applicable?	HT/GHT Role (one phrase)	DL Alternative	HT Better / Worse / Same?
S1 IMAGE INPUT	NO	Requires edge/boundary features	CNN preprocessing	WORSE
S2 PRE-PROCESSING	NO	Uses edge map as input	CNN denoising	WORSE
S3 FEATURE EXTRACTION	YES	R-table / boundary features	CNN feature extractor	WORSE
S4 DETECTION	YES	Parameter-space voting / peaks	YOLO	WORSE
S5 POST-PROCESSING	YES	Peak selection / thresholding	NMS / confidence filtering	COMPARABLE
S6 OUTPUT	NO	Outputs detection parameters	Prediction/output layer	COMPARABLE